

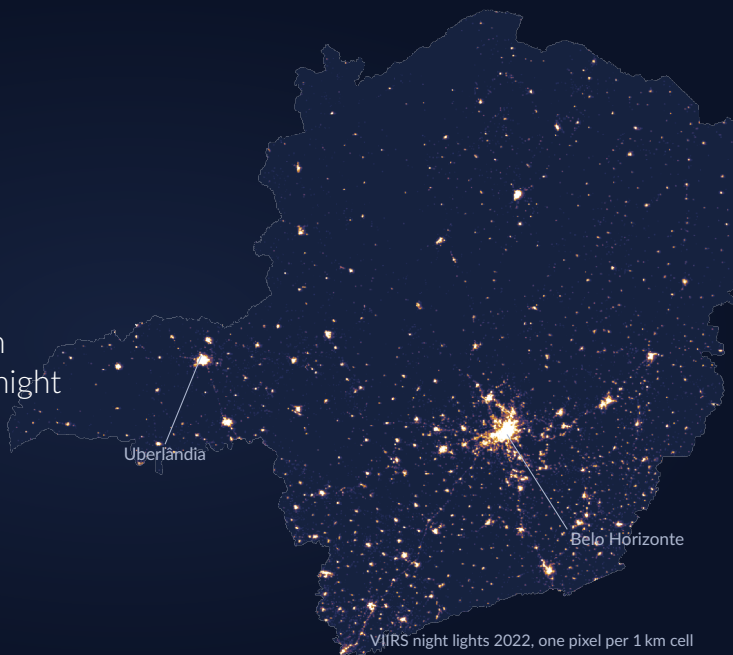
Reading the economy from space

Estimating municipal GDP from satellite embeddings, text and night lights

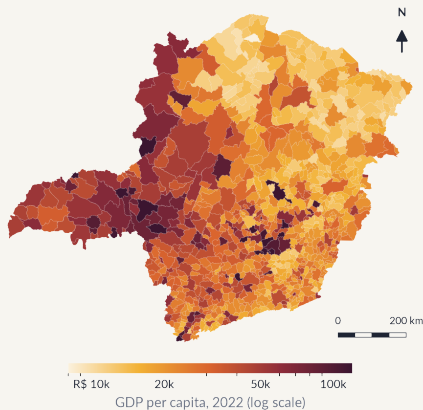
Hemant (22CS30029)

Minas Gerais, Brazil · 853 municipalities

October 2026



Local GDP is patchy; satellites see every place



49×

gap in GDP per capita inside one state:

Catas Altas (R\$ 391k) vs São João das Missões (R\$ 8k)

- Sub-national GDP is published **late**, **estimated** from partial data, or **missing** where statistics are weak.
- Satellites measure **every place, every year, with the same sensor**.

GOAL

Estimate GDP per capita for all **853 municipalities** of Minas Gerais from **free satellite data**, and test it **honestly**.

Two recent ideas, never tested together

UrbanCLIP Yan et al., WWW 2024

A language model writes a **caption for every satellite tile**; image and text are aligned.

- GDP gain only +0.5% R^2 ; random split lets neighbours leak into the test

Satellite embeddings Yue, Zhao & Hu, 2026

Google's **64-number embeddings, steered by night lights** towards economic meaning.

- countries only; no text, no POI or roads

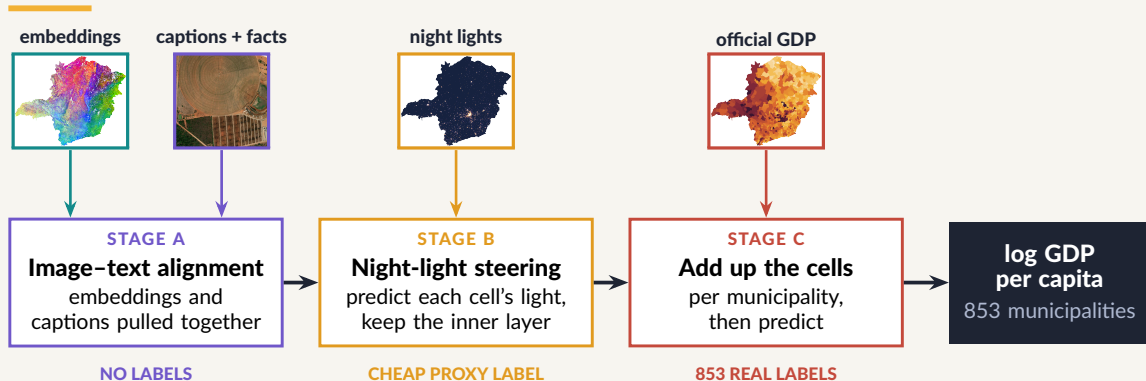
THIS PROJECT

Both, in one pipeline

- embeddings ● text
- night lights ● POI, roads

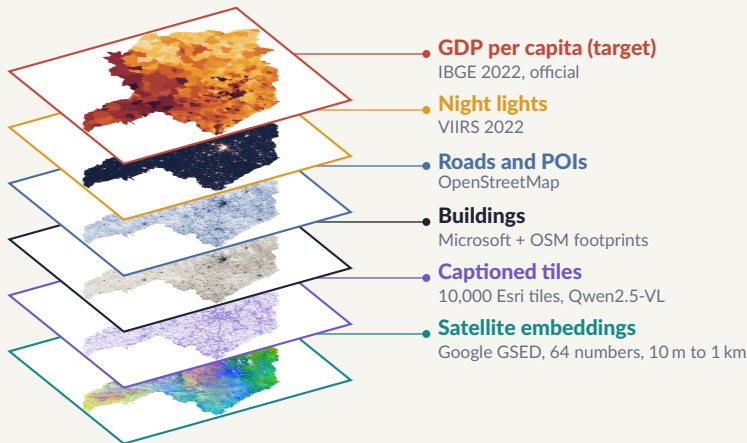
853 municipalities · spatial test
ablation of every ingredient

Three learning stages, from no labels to official GDP



Captions are needed only in training: at prediction time each cell uses its embedding alone. Switching stages off gives the ablations (variants V3-V9).

Six free layers on one 1 km grid



853

municipalities

575k

1 km cells

10,000

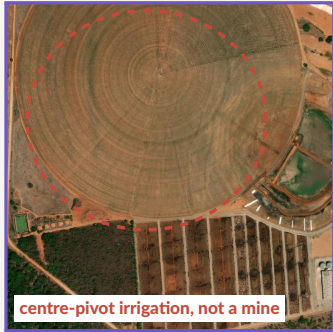
captioned tiles

10M

building footprints

All layers free and aligned on the same grid (EPSG:31983). The target is official statistics, not night lights.

An AI model describes each tile; map data vetoes false claims



QWEN2.5-VL-7B CAPTION, AFTER THE CHECKS

- ✗ The satellite image shows a large circular area with concentric rings, likely a mine pit, surrounded by a grid of smaller rectangular plots of farmland: **no industry in the data**
- ✓ Roads crisscross the area, connecting the plots and the perimeter.
- ✓ Vegetation is sparse, mostly along the edges and in some patches within the farmland.
- ✓ A small body of water, possibly a pond, is visible near the farmland.
- ✓ No buildings or industrial structures are evident, suggesting minimal built-up area.
- ✗ The overall scene appears to be an agricultural landscape with a focus on farming and possibly mining operations: **filler**

Esri World Imagery, about 1 m per pixel

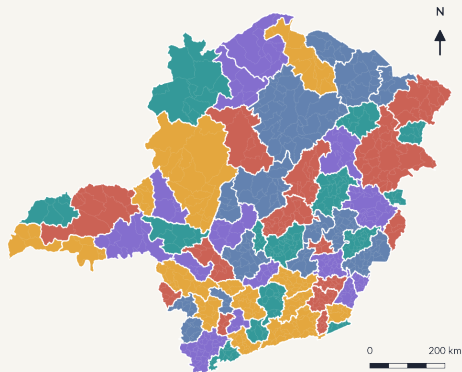
SENTENCES REMOVED, BY RULE

Filler, no content	4,183
Industry / mine	1,130
Building	1,108
Water	1,046
Road	1,039
Farmland	732

84% kept of 60,023 sentences

Checked against ESA land cover, OSM roads and building footprints; image-text matching removes 246 more captions. A blind hand audit is in progress.

Every municipality is tested in a region never seen



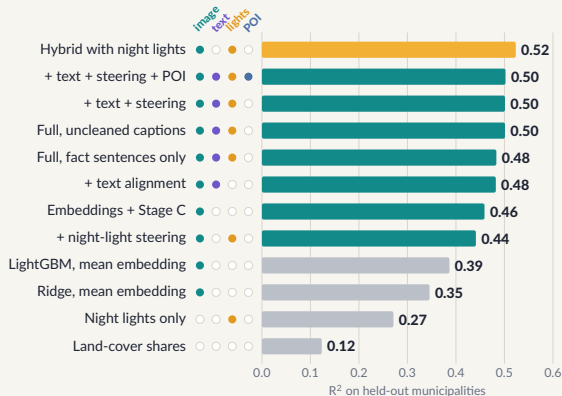
70 official regions (white outlines) grouped into 5 folds

	F1	F2	F3	F4	F5
Round 1	test	validate	train	train	train
Round 2	train	test	validate	train	train
Round 3	train	train	test	validate	train
Round 4	train	train	train	test	validate
Round 5	validate	train	train	train	test
	train	validate	test		

- Whole regions held out: **no look-alike neighbour** in training. All stages retrained each round.
- UrbanCLIP split tiles **at random**, so neighbours sat on both sides and scores were inflated.

Models are compared on the same folds with a block bootstrap over regions (95% intervals).

Text gives the biggest gain; the hybrid explains half of GDP



Dots: ingredients each model uses. Hybrid = best model blended with night lights.

0.52 R^2 , hybrid, unseen regions

RQ2 TEXT largest gain, borderline

$\Delta RMSE$ -0.025, 95% CI [-0.050, +0.004]

RQ1 STEERING no clear effect

$\Delta RMSE$ +0.007, 95% CI [-0.028, +0.041]

RQ3 POI, ROADS nothing added

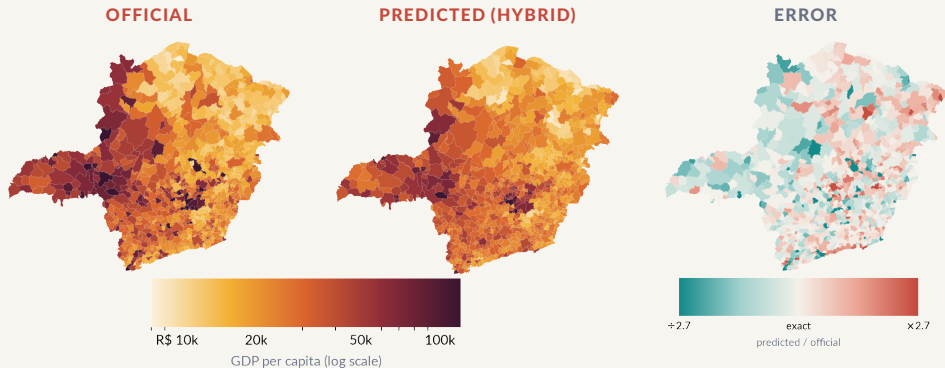
$\Delta RMSE$ -0.001, 95% CI [-0.017, +0.016]

RQ4 HYBRID best, not significant

$\Delta RMSE$ -0.009, 95% CI [-0.035, +0.015]

RMSE differences on log GDP per capita with 95% block-bootstrap intervals; negative = first model better.

Predictions follow the map, but are pulled towards the middle



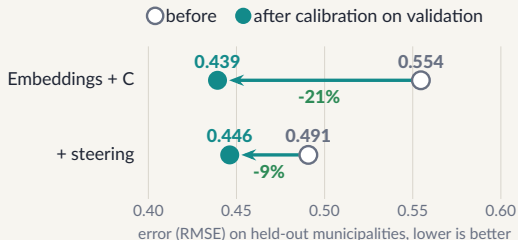
75%

of municipalities predicted
within a factor 1.5 of official

Under-predicted: rich agribusiness and mining towns (west, centre).

Over-predicted: the poorer north-east. Explaining these errors is RQ5, next.

Calibration mattered more than night-light steering



- 1 Predictions were over-spread.** A straight-line correction fitted on validation municipalities cut error by **-21%**.
- 2 Honest testing changes conclusions.** Steering looked helpful before the fix; after it, the gain vanished.
- 3 Captions need checking.** The model invents mines, roads and farmland on plain tiles; the data checks remove most of it.

Status

Data, grid and folds

 done

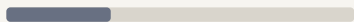
Captions and cleaning

 audit left

Models and experiments

 done (V1-V12)

Error analysis and maps

 code ready

RESEARCH QUESTIONS

RQ1 night-light steering 

RQ2 text (captions) 

RQ3 POIs and roads 

RQ4 hybrid 

RQ5 where it fails 

NEXT

caption audit → error analysis and 1 km maps (RQ5) → report

SO FAR

Satellite embeddings, captions and night lights, tested honestly, explain half of local GDP.

0.52

R^2 , hybrid, unseen regions

0.27

R^2 , night lights alone

—21% 84%

error after calibration

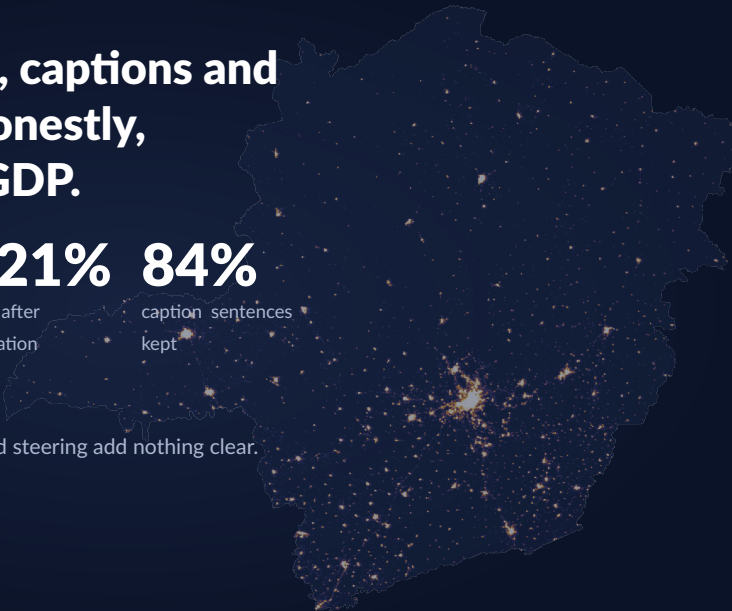
caption sentences kept

So far: captions give the largest gain, POIs and steering add nothing clear.

Next: where does the model fail (RQ5)?

Questions?

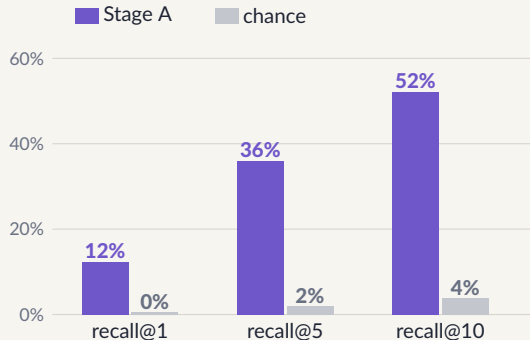
github.com/chadsaras/GIS_Project



Data sources

Layer	Source	Use
● Satellite embeddings	Google Satellite Embedding V1 (AlphaEarth), 2022	image input, averaged 10 m to 1 km
● Night lights	VIIRS DNB annual V2.2, 2022	Stage B proxy label, night-light benchmark
● Land cover	ESA WorldCover v200, 2021	masking, fact sentences, caption checks
● Roads, POIs	OpenStreetMap (Geofabrik, 2023-01-01)	features, fact sentences, caption checks
● Buildings	Microsoft ML footprints + OSM	caption checks
● Caption imagery	Esri World Imagery (ArcGIS Location Platform)	input to the vision-language model
● GDP, population	IBGE municipal GDP 2022, Census 2022	target: log GDP per capita

Stage A aligns embeddings and captions far above chance



For a validation tile, how often is its own caption among the top k of 256 candidates?

Part of the alignment comes from the fact sentences; variants V8 (raw captions) and V9 (facts only) separate the two.

What the captions look like



FOREST

A predominantly forested area with patches of cleared land.



FARMLAND

A large, circular area of brown farmland, likely recently plowed, dominating the center.



VILLAGE EDGE

A sparsely built area with scattered houses and small buildings, likely residential.



TOWN CENTRE

A densely built-up area with numerous houses featuring red-tiled roofs, indicating a residential neighborhood.

First sentence of each caption that passed the checks.